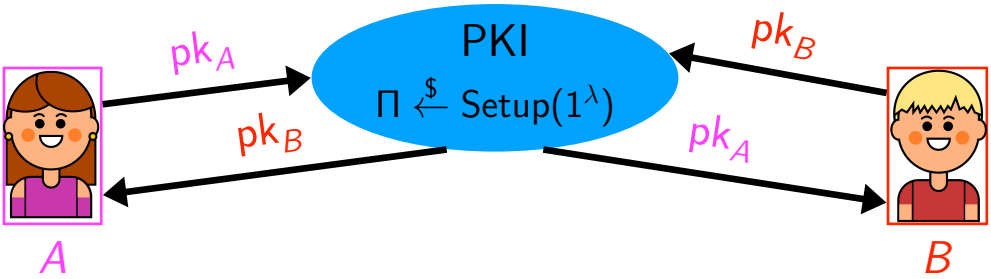
# Lattice-based NIKE with optimal tightness

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# NIKE



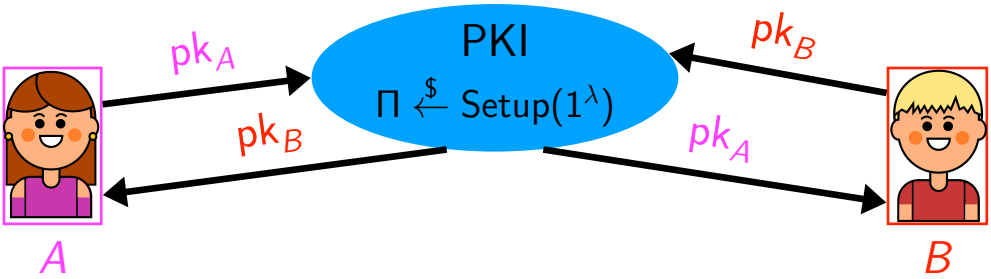

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$K_{A,B} \leftarrow \mathsf{SharedKey}(\Pi, \mathsf{sk}_A, \mathsf{pk}_B)$

$(\mathsf{sk}_B, \mathsf{pk}_B) \xleftarrow{\$} \mathsf{KeyGen}(\Pi)$

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# NIKE




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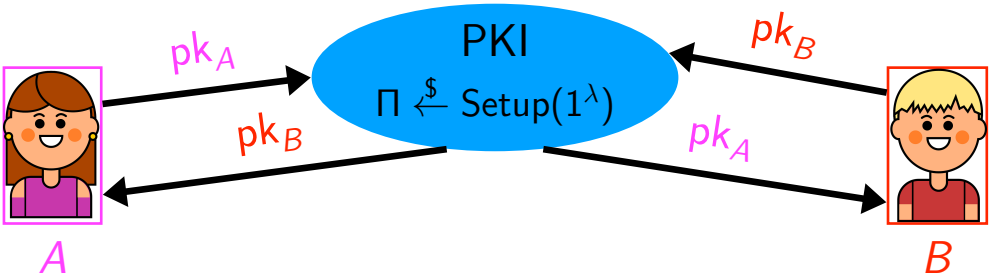
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Perfect correctness: $K_{A,B} = K_{B,A} \neq \perp$

# NIKE




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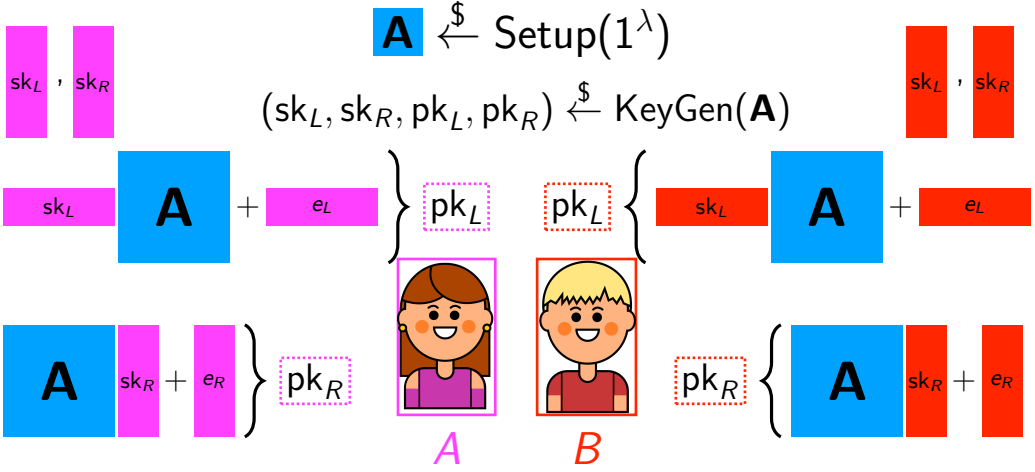
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$K_{B,A} \leftarrow \mathsf{SharedKey}(\Pi, \mathsf{sk}_B, \mathsf{pk}_A)$

Non-perfect correctness: $\Pr[K_{A,B} = K_{B,A} \neq \perp] \geqslant 1 - \epsilon_{\mathsf{corr}}, \epsilon_{\mathsf{corr}} \in (0, 1]$

# Lattice-based NIKE [folklore]



# Lattice-based NIKE



$$\mathsf{sk}_L \cdot \mathsf{pk}_R + e'_L = \mathsf{sk}_L \cdot \left( \mathbf{A} \cdot \mathsf{sk}_R + e_R \right) + e'_L \approx$$

$A \quad B$

$$\mathsf{pk}_L \cdot \mathsf{sk}_R + e'_R = \left( \mathsf{sk}_L \cdot \mathbf{A} + e_L \right) \cdot \mathsf{sk}_R + e'_R$$



$\mathsf{sk}_L \cdot e_R + e'_L$, $e_L \cdot \mathsf{sk}_R + e'_R$ are short

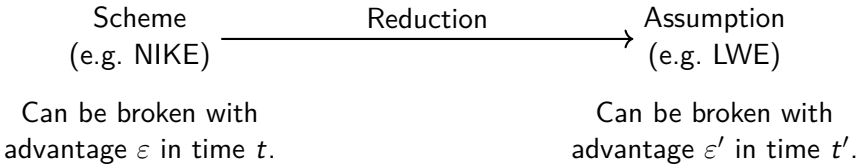
LWE assumption: $\left( \mathbf{A}, \mathsf{sk}_L \cdot \mathbf{A} + e_L \right) \approx_c \left( \mathbf{A}, \mathsf{pk}_L \right)$

# Security model

- Adaptive honest key registration (HKR) security:
  - Many users can be registered
  - Adaptive corruption of users (obtain sk)
  - Challenge: Distinguish real shared key (from uncorrupted user pair) from random key

# Tightness

- Security of scheme: reduction to well-known hardness assumption




- Security loss $\ell$: gap between hardness guarantee of scheme and assumption

$$\frac{t'}{\varepsilon'} \leqslant \ell \cdot \frac{t}{\varepsilon}$$

$\Rightarrow$ **Compensate loss** with larger system parameters $\Rightarrow$ **Less efficient** scheme

# Related works on tightness of NIKE

Setting: Two-party NIKE, $n$ registered parties

| NIKE class | Security Loss |
|---|---|
| **Unique shared key – Bader, Jager, Li, Schäge [BJLS16]** | $\Omega(n^2)$ |
| **Weakly unique shared key – Hesse, Hofheinz and Kohl [HHK18]** | $\Omega(n)$ |
| **Inner-product structure – Hesse, Hofheinz, Kohl and Langrehr [HHKL21]** | $\Omega(n/\lvert\mathsf{pk}\rvert)$ |

# Related works on tightness of NIKE

Setting: Two-party NIKE, $n$ registered parties

| NIKE class | Security Loss | Applicable to lattice-based NIKE? |
|---|---|---|
| **Unique shared key – Bader, Jager, Li, Schäge [BJLS16]** | $\Omega(n^2)$ | **No:** No unique shared key Non-perfect correctness |
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| **Inner-product structure – Hesse, Hofheinz, Kohl and Langrehr [HHKL21]** | $\Omega(n/\lvert\mathsf{pk}\rvert)$ | **No:** No inner product structure Non-perfect correctness |

# Lattice-based NIKE

- Naturally achievable security loss: $O(n^2)$

| scheme | security loss |
|---|---|
| folklore | $O(n^2)$ |
| [Lan23] | $O(n^2)$ |
| SWOOSH [GdKQ+24] | $O(n^2 q_H^2)$ |

# Lattice-based NIKE

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- Can we do better? There was no lower bound applicable to lattice-based NIKE.

Optimal security loss for lattice-based NIKE remained an open question.

# Our results

| scheme | security loss |
|---|---|
| folklore | $O(n^2)$ |
| [Lan23] | $O(n^2)$ |
| SWOOSH [GdKQ+24] | $O(n^2 q_H^2)$ |
| Ours | $O(n)$ |

- Can we do better w.r.t. security loss? **Yes!**
  - **construction of lattice-NIKE scheme with security loss $O(n)$**
  - **matching lower bound $\Omega(n)$**
- **Fully closed the research gap**

How did we do that? **Pairwise checkability property**

# Pairwise checkability property

| | |
|---|---|
| **Bader et al. [BJLS16]** | $\mathsf{Valid}(\mathsf{pk}_A) \wedge \mathsf{Valid}(\mathsf{pk}_B)$ $\rightarrow$ shared key fixed by $(\mathsf{pk}_A, \mathsf{pk}_B)$ |
| **Hesse, Hofheinz, Kohl [HHK18]** | $\mathsf{Valid}(\mathsf{pk}_A, \mathsf{sk}_A) \wedge \mathsf{Valid}(\mathsf{pk}_B, \mathsf{sk}_B)$ $\rightarrow \mathsf{SharedKey}(\mathsf{sk}_A, \mathsf{pk}_B) = \mathsf{SharedKey}(\mathsf{sk}_B, \mathsf{pk}_A)$ |
| **Our work (pairwise checkability)** | $\mathsf{Valid}(\mathsf{pk}_A, \mathsf{sk}_A, \mathsf{pk}_B) \wedge \mathsf{Valid}(\mathsf{pk}_B, \mathsf{sk}_B, \mathsf{pk}_A)$ $\rightarrow \mathsf{SharedKey}(\mathsf{sk}_A, \mathsf{pk}_B) = \mathsf{SharedKey}(\mathsf{sk}_B, \mathsf{pk}_A)$ |

- Pairwise checkability is **weaker** than weakly unique shared key property [HHK18]
  - **Generalizes** result for NIKE classes with [weakly] unique shared key property
  - **Satisfied** by lattice-based NIKE schemes

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The optimal security loss for lattice-based NIKE is $\Theta(n)$.

# References I




Christoph Bader, Tibor Jager, Yong Li, and Sven Schäge.
On the impossibility of tight cryptographic reductions.
In Marc Fischlin and Jean-Sébastien Coron, editors, EUROCRYPT 2016, Part II, volume 9666 of LNCS, pages 273–304. Springer, Berlin, Heidelberg, May 2016.
doi:10.1007/978-3-662-49896-5_10.



Phillip Gajland, Bor de Kock, Miguel Quaresma, Giulio Malavolta, and Peter Schwabe.
SWOOSH: Efficient lattice-based non-interactive key exchange.
In Davide Balzarotti and Wenyuan Xu, editors, USENIX Security 2024. USENIX Association, August 2024.
URL:
https://www.usenix.org/conference/usenixsecurity24/presentation/gajland.

# References II



Julia Hesse, Dennis Hofheinz, and Lisa Kohl.
On tightly secure non-interactive key exchange.
In Hovav Shacham and Alexandra Boldyreva, editors, CRYPTO 2018, Part II, volume 10992 of LNCS, pages 65–94. Springer, Cham, August 2018.
doi:10.1007/978-3-319-96881-0_3.


Julia Hesse, Dennis Hofheinz, Lisa Kohl, and Roman Langrehr.
Towards tight adaptive security of non-interactive key exchange.
In Kobbi Nissim and Brent Waters, editors, TCC 2021, Part III, volume 13044 of LNCS, pages 286–316. Springer, Cham, November 2021.
doi:10.1007/978-3-030-90456-2_10.

# References III


Roman Langrehr.
On the multi-user security of LWE-based NIKE.
In Guy N. Rothblum and Hoeteck Wee, editors, TCC 2023, Part IV, volume 14372 of LNCS, pages 33–62. Springer, Cham, November / December 2023.
doi:10.1007/978-3-031-48624-1_2.